RapidBoxForest: Link-Envelope-Guided Box Forests for Low-Overhead Scene Construction in Multi-Query Manipulation

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***Abstract*—Repeated-query manipulation benefits from reusable planning structure, but certified convex regions can require costly construction and recertification in cluttered workcells. RapidBoxForest (RBF) checks axis-aligned joint boxes with endpoint-to-link swept-capsule bounds, organizes boxes accepted by the scene-validation policy into a scene-level forest and typed corridor graph, and stores kinematics-keyed envelope evidence in a Lifelong Envelope Cache Tree (LECT) separately from scene labels. Certificates are conditional: they apply only to conservative box unions or OBB-corridor unions (OBB: oriented bounding box); repaired or shortened paths, and paths using filter-only evidence, require independent query validation. Saved-catalog experiments use audit-separated timing and explicit cache accounting. On Shelf+IIWA, warm-cache depth-23 LECT replay builds the reusable scene structure in 0.166 s, with 0.130 s per-query online time and 0.164 s five-query amortized time; live hierarchical interval forward kinematics (HIFK-5) materialization takes 7.006 s. At the saved-catalog random-scene operating points, RBF has lower per-query online time than the Batch Informed Trees (BIT*) reference rows in all six robot–difficulty cases, while BIT* has lower median path ratios than RBF in the four UR5/Panda cases. In a maintenance-only study, leaf-sweep insertions are 153.3–207.5× faster and removals are 943.6–3354.0× faster than the 15-obstacle warm build. Snapshot creation, loading, and resident memory are excluded from the reported timing rows; update rows report maintenance only, not a complete pipeline that updates the scene and solves a query.**

***Index Terms*—Motion planning, multi-query planning, link interval envelopes, box-region planning, configuration-space planning.**

I. Introduction

Many high-degree-of-freedom manipulation workloads contain related queries in nearly fixed workcells. Robot kinematics persist, while start-goal pairs, local obstacles, or small scene edits change. Convex-region planners, including IRIS (Iterative Regional Inflation) and Graphs of Convex Sets (GCS), provide optimization-ready region-level abstractions once regions are certified [1], [2], [3], but region construction can be costly in high-dimensional or frequently edited scenes. Sampling-based planners, including PRM and RRT variants, avoid such region construction [4], [5], [6], [7], but their reusable structures are zero-volume roadmaps or query trees.

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RBF addresses this trade-off with a low-overhead volumetric model: it grows axis-aligned joint-space boxes, checks them with link-envelope bounds, connects policy-accepted boxes into a typed corridor graph, and reuses that graph across later queries. The resulting trade-off is explicit: boxes are less expressive than more general convex covers, but they simplify validation, graph building, updates, and multi-query reuse.

a) Problem statement: Let $\mathcal{Q} \subset \mathbb{R}^n$ be the joint-limit box of a fixed manipulator and let $\mathcal{O} \subset \mathbb{R}^3$ be the current workspace obstacle set. Given a set or stream of queries $(q_s, q_g) \in \mathcal{Q}^2$, RBF builds a finite family of policy-accepted configuration-space (C-space) boxes and an adjacency graph over those boxes. The design goal is low scene-construction and maintenance overhead while retaining reusable volumetric planning structure and conservative evidence where the stated validation policy supports it.

The geometric core adapts classical endpoint bounding. For rounded links, endpoint envelopes bound endpoint trajectories over a joint box; their radius-expanded convex hull encloses the swept link and can certify C-space boxes when the endpoint source and separation tests are conservative. Conservative sources yield conditional box certificates, whereas tighter non-certifying sources remain filters whose paths require query validation. Reuse is split accordingly: scene-level RBF caches store one scene's box forest and corridor graph, while the Lifelong Envelope Cache Tree (LECT) stores replayable kinematics-keyed endpoint and link evidence.

The evaluation follows the same boundary. Success is counted only after fixed-step query validation unless a row is explicitly conservative. Warm-cache Shelf rows include replay-based scene construction and per-query retrieval; snapshot creation, loading, resident memory, and offline setup amortization are excluded. Dynamic update rows measure scene-cache maintenance, not a complete pipeline that updates the scene and solves a query. Thus the experiments evaluate low-overhead reusable-region construction under a stated validation policy, not dominance over single-query planners, repair-free certification, or completeness. This work contributes:

1) *Box validation:* conservative endpoint-to-link envelopes validate boxes; non-conservative endpoint sources remain query-validated filters.
2) *Planning object:* policy-accepted boxes form a reusable scene forest and typed corridor graph.

TABLE I

Qualitative taxonomy of reusable manipulation-planning paradigms. The comparison emphasizes the persistent object, validation granularity, and scene-edit semantics rather than ranking planners by runtime.

Family	Persistent object	Vol.	Validation and certificate unit	Scene-edit reuse	Main trade-off
PRM and LazyPRM	Roadmap vertices and edges	No	Local edges or path segments	Topology can be reused; affected edges require rechecking	Reusable multi-query graph, but reusable evidence is zero-volume.
RRT-Connect and RRT	Query tree	No	Incremental path segments	Little persistent reuse across queries	Query-focused discovery; persistent structure is query-local.
BIT* and FMT*	Batch sample graph or tree	No	Candidate graph edges	Primarily query-centric; limited scene-level reuse	Anytime quality improvement, but still edge-centric.
Adaptive cell decomposition	Configuration-space cells	Yes	Cells under exact, sampled, or interval tests	Explicit cells can be locally relabeled when maintained	Volumetric representation; scalability depends on the cell validator.
IRIS and GCS	Certified convex regions and graph	Yes	Convex collision-free regions	Region reuse is scene-coupled; edits may require reinflation or recertification	Optimization-ready regions with construction and recertification cost.
RBF	Box forest + typed graph + LECT evidence	Yes	Joint boxes via link envelopes; query validation for filter-only paths	Scene boxes are maintained locally; LECT replays kinematics-keyed evidence	Low-overhead volumetric reuse with less expressive boxes and conditional certificates.

Abbrev.: PRM=probabilistic roadmap; RRT=rapidly exploring random tree; BIT*=Batch Informed Trees; FMT*=Fast Marching Tree; IRIS=Iterative Regional Inflation; GCS=Graphs of Convex Sets; LECT=Lifelong Envelope Cache Tree; Vol.=volumetric.

3) *Two-level reuse:* scene caches hold boxes and graph state; LECT replays kinematics-keyed envelope evidence.
4) *Local scene-cache maintenance:* deletions promote blocked domains; insertions trigger bounded refill.
5) *Evaluation protocol:* saved catalogs, audit-separated timing, and cache accounting define the evaluation scope.

Sections III–VI cover link envelopes, LECT, forest construction and updates, and evaluation.

II. Related Work

Convex-region planners construct collision-free convex regions and search or optimize on the induced graph. IRIS variants inflate collision-free convex regions, and Graphs of Convex Sets (GCS) make the resulting graph optimization explicit [1], [2], [3], [8], [9], [10]. RBF shares the region-level objective but uses axis-aligned C-space boxes; this restricted geometry limits expressivity while keeping validation, adjacency, and update operations local and explicit. Table I summarizes the qualitative position.

a) Continuous-collision-detection endpoint bounds: Swept-volume and interval-kinematics methods supply the geometric primitives used here. Convex-hull generator bounds and capsule radius lifts provide swept-set enclosures, while the Proximity Query Package (PQP), Gilbert–Johnson–Keerthi (GJK), the Flexible Collision Library (FCL), and interval Denavit–Hartenberg propagation provide collision-query and interval-kinematics primitives used in related continuous-collision-detection pipelines [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22]. Redon *et al.* apply this pipeline to articulated-body continuous collision detection over time steps [23], [24]; related Taylor-model and safety methods use similar swept-sphere bounds [25], [26], [27]. RBF uses joint-box intervals rather than trajectory time steps as the bounding domain, stores compatible interval evidence in LECT, and assembles policy-accepted boxes into a typed corridor graph. Repaired or post-processed paths remain subject to query validation.

b) Reuse and adaptive decomposition: Multi-query planners amortize effort through reusable graphs. Probabilistic roadmaps (PRMs) and LazyPRM reuse roadmaps and edge checks [6], [28], [29], [30], experience-based systems reuse prior paths [31], and adaptive cell methods split cells until the accepted-cell graph can be searched [32], [33]. RBF combines reusable graph search with box-level envelope validation and a separate kinematics-keyed evidence cache.

c) Sampling baselines and refinement: RRT*, RRT-Connect, Batch Informed Trees (BIT*), Fast Marching Tree (FMT*), and the Open Motion Planning Library (OMPL) provide standard single-query or anytime implementations and benchmarking infrastructure [7], [34], [35], [36], [37], [38], [39]. Trajectory optimizers refine supplied paths [40], [41], [42], [43]. RBF is complementary: it supplies reusable volumetric context before repair, smoothing, and query validation.

III. Link Interval Envelopes

The link-envelope validation layer maps joint regions to endpoint and link bounds. For a joint region $I \subseteq \mathcal{Q}$, endpoint envelopes $E_k(I)$ bound endpoint trajectories and link envelopes $B_\ell(I)$ bound rounded links over the same region:

$$I \mapsto \{E_k(I)\} \mapsto \{B_\ell(I)\}.$$

Axis-aligned joint boxes are the default regions; local oriented bounding boxes (OBBs) in scaled joint coordinates are used only for typed bridge corridors. Conservative endpoint evidence supports certificates only after link construction and scene-separation tests. Empirical or otherwise non-conservative endpoint sources are filters and must be followed by query validation.

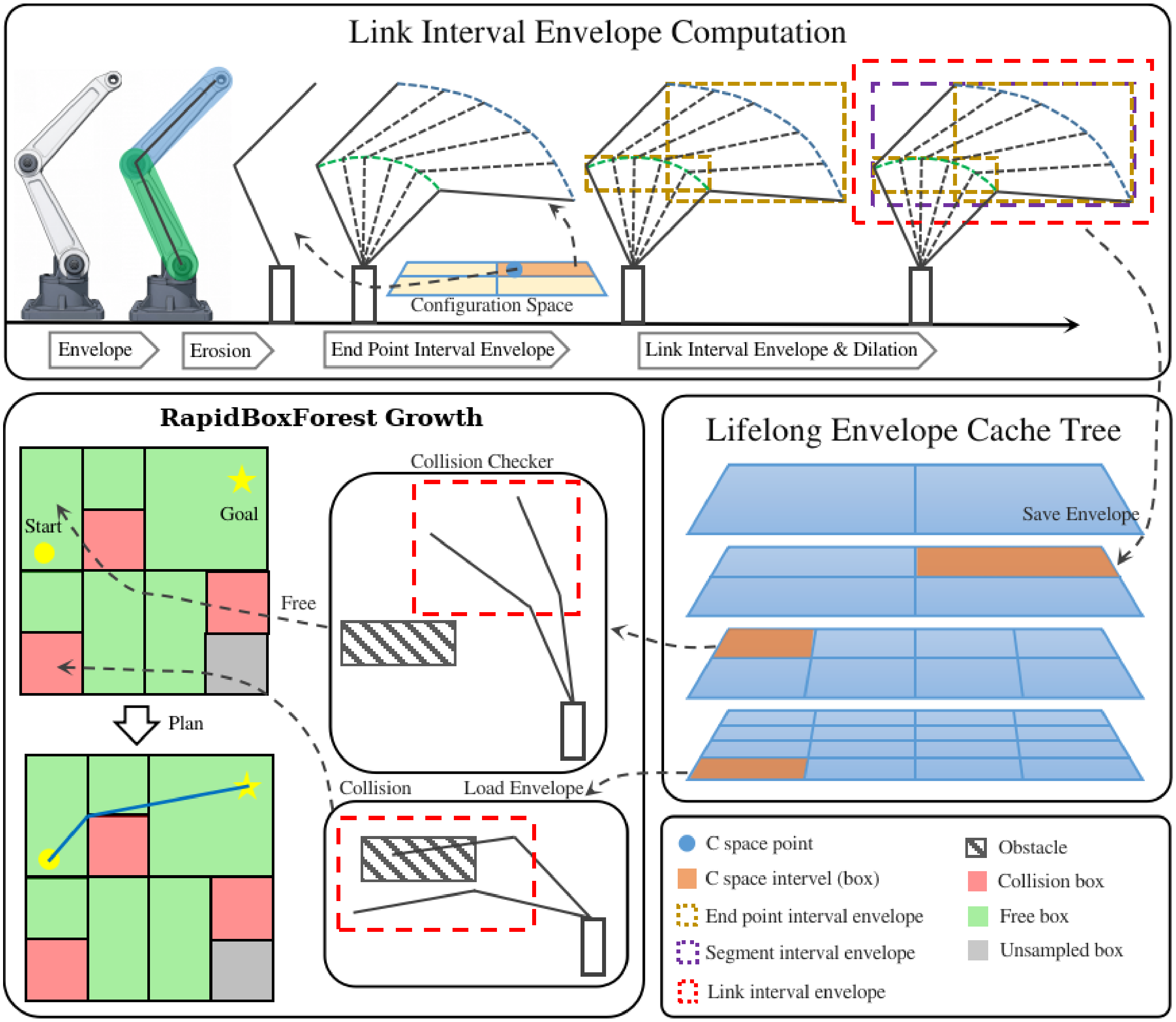


Fig. 1. RBF pipeline for link-envelope-guided multi-query planning. Top: endpoint interval envelopes induce link interval envelopes over a configuration-space (C-space) box. Middle: LECT reuses kinematics-keyed endpoint and link-envelope records across repeated builds. Bottom: scene filtering and sample-guided growth expand a policy-accepted scene-level box forest for repeated corridor queries.

Assumption 1 (Collision model for certificates)**.** *The certificate model covers active rounded links against $\mathcal{O}$, with joint limits enforced by $I \subseteq \mathcal{Q}$. Self-collision, attachments, grippers, task regions, and other constraints are outside this certificate unless encoded in the validation policy and cache descriptor; the experiments check only collisions between active links and workspace obstacles.*

A. Geometric Foundation

Let $T(q)x = R(q)x + t(q)$ be the transform at configuration q, and let $\mathcal{S}_I(C_0)$ denote the workspace set swept by body C_0 as q ranges over I. For a capsule $C = S \oplus B_2(r)$, the fixed radius ball commutes with rigid motion:

$$\mathcal{S}_I(C) = \mathcal{S}_I(S) \oplus B_2(r).$$

Thus a rounded link can be bounded by first bounding the zero-radius center segment and then lifting by the link radius. For a convex body $P = \mathrm{conv}(v_1, \ldots, v_m)$, outer bounds on the generator trajectories also bound the swept body,

$$\mathcal{S}_I(P) \subseteq \mathrm{conv}\Big(\bigcup_i \widehat{\Gamma}_i(I)\Big).$$

For a capsule link, only the two endpoint trajectories are needed. Let $\Gamma_a(I)$ and $\Gamma_b(I)$ denote the exact workspace trajectories of the proximal and distal endpoints over I. RBF uses these swept-sphere bounds over a joint-box domain and stores the resulting bounds as reusable kinematics-keyed bound records.

Proposition 1 (Endpoint envelopes induce link interval envelopes)**.** *Consider a rounded link $L = [a, b] \oplus B_2(r)$ and a joint interval I. If endpoint envelopes $E_a(I)$ and $E_b(I)$ satisfy $\Gamma_a(I) \subseteq E_a(I)$ and $\Gamma_b(I) \subseteq E_b(I)$, then*

$$\mathcal{S}_I(L) \subseteq \mathrm{conv}\big(E_a(I) \cup E_b(I)\big) \oplus B_2(r).$$

Consequently, any outer representation of the right-hand side is a valid link interval envelope for the whole joint box I.

Proof. For each $q \in I$, the center segment of the moved link is the convex hull of the two moved endpoints, so $T(q)[a, b] \subseteq \mathrm{conv}(\Gamma_a(I) \cup \Gamma_b(I))$ over the entire interval. The endpoint inclusions place this set inside $\mathrm{conv}(E_a(I) \cup E_b(I))$. Adding the fixed radius ball commutes with the rigid-motion sweep of the capsule, giving the stated containment. □

Corollary 1 (Conservative joint-box validation)**.** *Let $I \subseteq \mathcal{Q}$ be a joint box and let $\{B_\ell(I)\}$ be conservative link interval envelopes for all active rounded links. If $B_\ell(I) \cap \mathcal{O} = \emptyset$ for every link ℓ, then every configuration $q \in I$ is collision-free with respect to the collision model in Assumption 1.*

Proof. By Proposition 1, each conservative link envelope contains the swept workspace volume of its link over all configurations in I. Disjointness from $\mathcal{O}$ for every active link therefore excludes workspace obstacle collision for the whole manipulator at every $q \in I$ under Assumption 1. □

a) Numerical conservatism: A certificate is attached only when endpoint propagation, radius padding, and obstacle-disjointness tests are outward bounds under Assumption 1. Axis-aligned bounding box (AABB) padding uses a conservative B_∞ radius lift. SupportHull/GJK is certificate-backed only with outward obstacle inflation, margins, and termination tolerances. Empirical endpoint sources and post-processing outside a conservative box union or OBB-corridor union are treated as non-conservative filters; they do not extend Corollary 1 or Theorem 1.

B. Endpoint Interval Envelopes

For endpoint k, let $\Gamma_k(I)$ be its exact trajectory over interval I. Any conservative set containing $\Gamma_k(I)$ is an endpoint interval envelope; the basic record is an endpoint AABB because it is compact and directly reusable by all link-envelope representations. The planner uses two certificate-backed endpoint sources. Interval forward kinematics (IFK) propagates affine-arithmetic bounds along the Denavit–Hartenberg chain; shared noise symbols preserve some trigonometric correlation, making it a low-cost certificate-backed endpoint source in this implementation. Hierarchical IFK (HIFK) bisects the joint interval and forms hulls of child endpoint boxes, reducing wrapping at higher materialization cost. IFK is the unsplit affine-arithmetic pass; HIFK is a hierarchical split-and-hull evaluation (for example, HIFK-5 means five split levels), so the split depth is part of the cache descriptor and cannot be exchanged with an unsplit record. A critical-sample source probes interval endpoints, midpoints, near-boundary configurations, and trigonometric turning candidates; it can reduce over-approximation relative to IFK, but does not certify coupled extrema. An analytical extrema source solves low-dimensional sinusoidal boundary subproblems, reducing one- and two-joint boundary cases to structured root search. It becomes conservative only with an interval-closure wrapper; the Analytical row in Table V is otherwise a diagnostic reference. The critical-sample and analytical sources serve as diagnostics or filters: any path using either source is accepted only after fixed-step query validation unless wrapped by an independent conservative certificate. During LECT tree descent, transient prefix transforms are reused so unaffected kinematic-chain prefixes are not recomputed; this is a runtime materialization optimization rather than persistent scene state. Fig. 1 shows the endpoint-to-link pipeline, LECT reuse, and box-forest use.

C. C-space OBB Endpoint Envelopes

Axis-aligned boxes remain the global forest primitive. Bridge repair motions can introduce joint correlations. For transitions with certificate support, RBF uses a validated local typed C-space OBB-corridor region while leaving the dyadic forest unchanged. OBBs are defined in scaled joint coordinates. Let $y = S(q - q_{\mathrm{ref}})$, where S is a diagonal joint scaling, and let

$$R_{\mathrm{obb}} = \{\, q_c + A\xi \mid \xi \in \Xi \,\},$$
$$\Xi = \prod_{i=1}^{m} [-1, 1], \qquad A = S^{-1} U \,\mathrm{diag}(h),$$

with U orthonormal in the scaled metric and h the halfwidth vector. The object R_{obb} is a configuration-space region, not a workspace OBB: the certificate comes from endpoint and link envelopes over R_{obb}, not from the fitted orientation.

a) Cover certificate: The conservative cover test tiles the local cube, maps each tile to joint space, and validates the resulting AABB hulls:

$$I_r = \mathrm{bbox}_q(q_c + AX_r).$$

If every I_r passes the standard conservative link-envelope scene test, then

$$R_{\mathrm{obb}} \subseteq \bigcup_r I_r \subseteq \mathcal{C}_{\mathrm{free}},$$

where $\mathcal{C}_{\mathrm{free}}$ is the collision-free configuration set under Assumption 1. When this test succeeds, the stored OBB certificate is a finite cover of ordinary box certificates.

b) Correlated endpoint envelope: A tighter source keeps the local OBB variables ξ correlated through the forward-kinematics computation:

$$q_j = q_{c,j} + a_j^\top \xi, \qquad \xi_i \in [-1, 1],$$

where $a_j^\top$ is the j-th row of A. Affine-arithmetic propagation then produces endpoint records

$$E_k(R_{\mathrm{obb}}) = c_k + \sum_i g_{k,i}\epsilon_i \oplus [-\rho_k, \rho_k].$$

Equivalently,

$$E_k(R_{\mathrm{obb}}) = p_k(q_c) + J_k(q_c)A\Xi \oplus [-\rho_k, \rho_k],$$

where each coordinate remainder is bounded conservatively, for example by

$$\rho_{k,\mu} \geq \frac{1}{2}\sum_{a,b}\left|\left(A^\top H_{k,\mu}A\right)_{ab}\right|,$$

with $H_{k,\mu}$ an interval or otherwise conservative Hessian bound over R_{obb}. If these remainders are sampled rather than outward bounded, the OBB source is a non-conservative filter and not a certificate.

OBB endpoint records feed the same endpoint-to-link construction as joint boxes. For a zonotope-plus-box endpoint record,

$$h_{E_k}(d) = d^\top c_k + \sum_i |d^\top g_{k,i}| + \sum_{\mu=1}^{3} |d_\mu|\rho_{k,\mu}.$$

The zero-radius link centerline support is

$$h_\ell^\circ(d) = \max\{h_{E_a}(d), h_{E_b}(d)\},$$

and the rounded-link envelope adds the capsule radius,

$$h_\ell(d) = h_\ell^\circ(d) + r_\ell\|d\|_2.$$

Coordinate-axis queries of h_ℓ provide an AABB broadphase, while the same support record can be used by the SupportHull/GJK narrow phase. An OBB-corridor edge is certificate-supported only under conservative endpoint bounds and separation tests; otherwise it remains a typed transition requiring query validation. The corridor layer is separate from the persistent dyadic LECT tree unless promoted to a scene-local graph layer.

D. Link-Envelope Representations

Endpoint records can be exported to multiple link representations without changing the upstream kinematic source. For a capsule link, $B_\ell(I) = B_\ell^\circ(I) \oplus \mathcal{R}(r_\ell)$, with $B_\ell^\circ(I)$ bounding the centerline sweep and $\mathcal{R}$ the radius lift.

a) AABB envelope: AABB envelopes wrap the two endpoint envelopes in one box and pad by the link radius:

$$B_\ell(I) = \mathrm{bbox}\big(E_{\ell-1}(I) \cup E_\ell(I)\big) \oplus B_\infty(r_\ell).$$

Segment subdivision can retain a union of padded sub-envelopes for more selective filtering, but the union must remain explicit to preserve the gain.

b) Fixed-direction slab envelope: A discrete oriented polytope record with k directions (k-DOP) stores lower and upper support values along a fixed direction set. Endpoint boxes are projected onto those directions, padded by the link radius, and merged over retained segment pieces. In the runtime pipeline, these slab tests form a low-cost intermediate filter: after the padded AABB broadphase, non-axis directions can reject close obstacle candidates before GJK. They are filters unless their endpoint source and separation tests are conservative.

c) SupportHull envelope: SupportHull keeps the two endpoint AABBs and link radius as a compact support record. For direction d, the support point is selected from the proximal or distal endpoint box, and obstacle refinement uses a fixed-size GJK narrow phase with outward obstacle inflation. This preserves more directional structure than a single AABB while keeping the cached record compact. When present, fixed-direction slab records act only as an intermediate filter between the AABB broadphase and SupportHull/GJK refinement; they do not change the certificate condition, which still depends on conservative endpoint evidence and outward scene tests.

IV. Lifelong Envelope Cache Tree

LECT is a replay layer below the scene-level RBF cache. It stores scene-independent endpoint and link-envelope evidence keyed by robot kinematics and joint intervals. Obstacle tests, box labels, graph connectivity, and query outcomes remain scene-local; every replayed envelope must still be checked against the current obstacle set.

A. Cache Key and Split Policy

LECT is a dynamically grown binary k-d tree over $\mathcal{Q}$. A node stores its joint interval, split metadata, cached endpoint evidence, and runtime residency state. The central invariant is that evidence is *kinematics-keyed* rather than *scene-keyed*: replay imports geometry bounds, never a scene-validation label.

a) Evidence/label separation: An envelope record is replay-compatible only when the robot model, endpoint source, envelope representation, split descriptor, link radii, and active-link set match the current build. The split policy is also part of this descriptor; alternative policies are separate cache settings. For a parent interval I, let I_d^- and I_d^+ be the two children obtained by splitting joint dimension d at its midpoint. Define the cumulative downstream envelope volume as

$$\mathcal{V}(I) := \sum_\ell \mathrm{vol}(B_\ell(I)).$$

One implemented schedule chooses depth-wise split dimensions that minimize the worst-child cumulative envelope volume,

$$d^* = \arg\min_d \max\left\{\mathcal{V}(I_d^-), \mathcal{V}(I_d^+)\right\}.$$

Depth-synchronous schedules keep replay deterministic; if the scheduled dimension is unavailable at a node, the runtime falls back to the widest valid dimension. Candidate dimensions are filtered by active-link reachability: splitting a joint that cannot affect any retained endpoint cannot tighten the downstream envelopes and is skipped for that descriptor.

b) Sparse interval staging: Deep cells need not materialize every ancestor. A planning cell can carry its exact joint interval, split counts, and descriptor fingerprint; if an exact replay record is absent, the interval is materialized on demand and then validated against the scene. Scene labels remain only in the forest or partition layer.

B. Runtime Use and Validation Boundary

LECT serves local box materialization; graph search uses the scene forest. Given a joint interval, the runtime replays or materializes endpoint evidence, derives the requested link representation, and applies the current scene-validation policy. In the evaluated pipeline, AABB broadphase precedes SupportHull/GJK refinement. Parent or child records can accelerate materialization, but replayed evidence alone is not a collision-free certificate. When a node is split, child-derived hulls can tighten an ancestor envelope; the reconstructed record remains representation-dependent and must be checked again against $\mathcal{O}$. Directional records are staged filters: a joint interval first passes the padded AABB broadphase, and only the remaining close obstacle candidates reach slab or SupportHull/GJK refinement. If no exact replay record exists, the interval is materialized on demand and then handled by the same validation path, so a cache miss changes cost but not semantics.

C. Storage and Warm-Build Semantics

Persistence is selective. Endpoint records and derived link-envelope records are the reusable evidence units. IFK+AABB records can be regenerated on demand, whereas slab or SupportHull records are stored only when their additional rejection power justifies lookup and storage cost. The prewarm pass fills target leaf records and propagates child hulls to reconstruct ancestors. This avoids recomputing direct forward kinematics independently at every ancestor depth. In a warm build, a cache hit reconstructs the requested link envelopes and revalidates them against the current obstacles. Warm builds still regrow scene-specific boxes; persisted box labels are not imported as certificates. The registered Shelf replay is read-only during timing, so the replay store does not accumulate new evidence during those measurements. Random-scene cache growth is treated separately as part of the per-robot profile.

V. RapidBoxForest Construction

Fig. 2 illustrates the planning layer. RBF builds a scene-specific set of policy-accepted joint boxes and stores their adjacency as a typed corridor graph. This pair is the scene cache: while the obstacle set is unchanged, later queries use box membership and graph search instead of rebuilding the cover. LECT supplies reusable kinematics-keyed evidence, but every policy-accepted scene box is still checked against the current obstacles.

A box is an axis-aligned joint interval $B = [l^{(1)}, u^{(1)}] \times \cdots \times [l^{(n)}, u^{(n)}] \subseteq \mathcal{Q}$. A *policy-accepted box* is policy-relative: it has passed the configured scene-validation policy. Under conservative link-envelope disjointness, this acceptance is a collision-free certificate; under a non-conservative filter, the box is only a planning region, and any path segment relying on it must pass query validation. Cells that do not pass validation are not inserted into the forest; selected blocked or inconclusive domains are retained for local refinement or later update.

A. Construction Objective

Given the joint-limit box $\mathcal{Q}$ and obstacle set $\mathcal{O}$, the build stage constructs

$$\mathcal{F} = \{B_1, \ldots, B_N\}, \qquad B_i \subseteq \mathcal{Q},$$

where each B_i satisfies the scene-validation policy. This budgeted set is concentrated around task anchors, query-relevant frontiers, and connectors. The reusable output is the pair $(\mathcal{F}, G)$, where G records adjacency and auxiliary transitions.

For boxes

$$B_i = \prod_{d=1}^{n} [l_i^{(d)}, u_i^{(d)}], \qquad B_j = \prod_{d=1}^{n} [l_j^{(d)}, u_j^{(d)}],$$

RBF treats them as connected when every coordinate overlaps or is separated by at most a small tolerance ε_c:

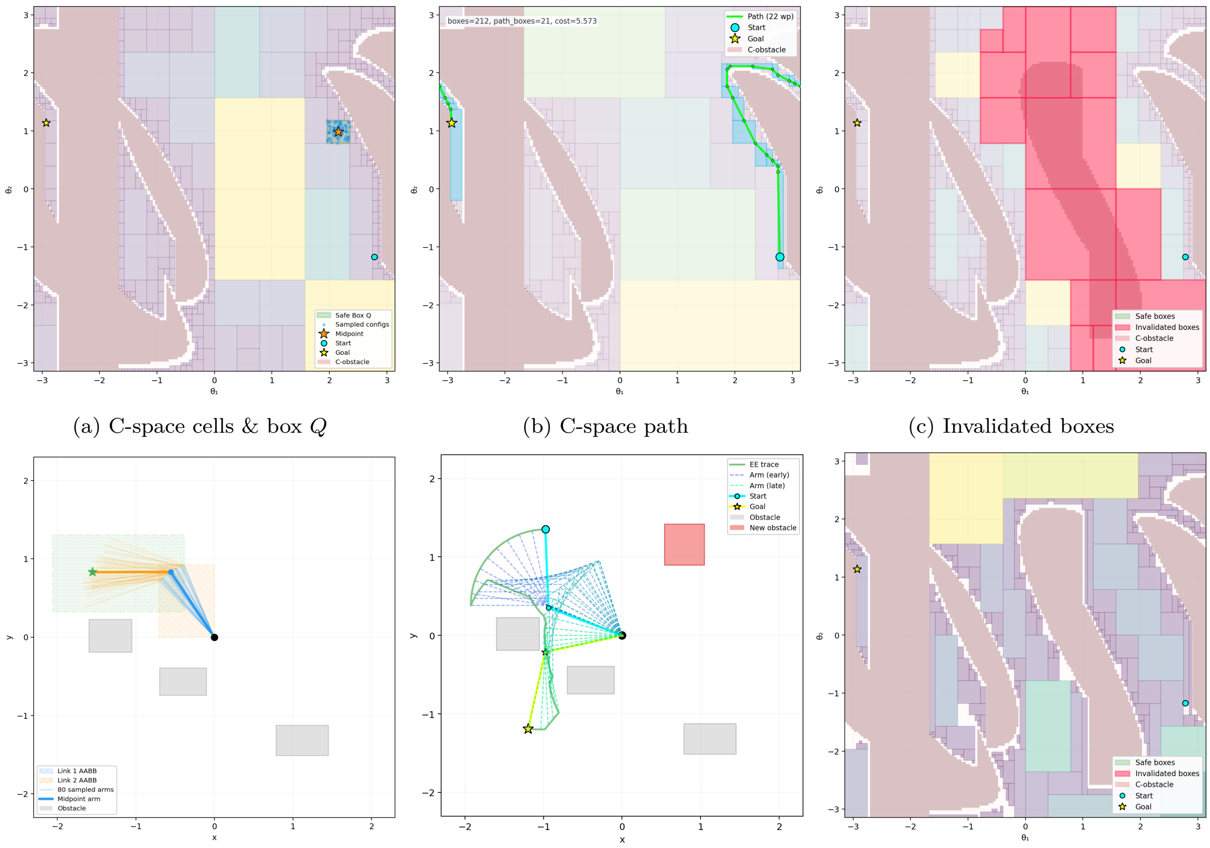
$$\max\{l_i^{(d)}, l_j^{(d)}\} \le \min\{u_i^{(d)}, u_j^{(d)}\} + \varepsilon_c, \qquad d = 1, \ldots, n.$$

Only exact overlap or contact directly yields a contained box corridor. Tolerance-gap and explicit segment edges are graph-search aids, not certificates, unless they are converted into conservative box-corridor or OBB-corridor regions; otherwise they require query validation.

B. Build Pipeline

The build proceeds through four stages. First, local materialization descends LECT around a seed and returns the first interval on that descent accepted by the scene-validation policy. Second, a leaf sweep classifies shallow cells and records blocked domains for possible refinement or later edits. Third, frontier growth adds boxes near anchors and query-relevant boundaries. Finally, consolidation merges or removes redundant boxes only when the resulting boxes still satisfy the same validation rule. Algorithms 1 and 2 define the two validation routines reused throughout this pipeline; frontier growth and consolidation repeat the same materialize–validate–connect pattern under budget limits and do not introduce a separate validation primitive.

a) Implementation controls: Validation semantics and timings are governed by the build budget, LECT descriptor, endpoint and link representation, admitted edge types, and audit resolution. Materialization failures remain deferred domains for restricted refinement; frontier or repair witnesses are certificate-supported only after conversion into conservative box-corridor or OBB-corridor regions. Lower-level scheduling constants are fixed in the artifact manifest.



(a) C-space cells & box Q (b) C-space path (c) Invalidated boxes

(d) Workspace AABBs of Q (e) Workspace path (f) After regrowth

Fig. 2. Two-link schematic of the RBF build, query, and obstacle-update cycle. A highlighted C-space box maps to workspace AABBs; the box corridor supplies a candidate path; obstacle insertion invalidates intersecting boxes and triggers local refill without a full rebuild. The panels illustrate the build, query, and update logic rather than the robot geometry, workspace scale, or dimensionality of the experiments.

Algorithm 1 Local box materialization around a seed.

Input: Seed q, LECT T, obstacle set $\mathcal{O}$, depth cap $D_{\max}$
Output: Policy-accepted interval or failure

start from the root interval containing q
while depth $\leq D_{\max}$ **do**
  replay or materialize current endpoint and link evidence
  if the current interval passes the scene-validation policy **then**
    return current interval
  end if
  split and descend to the child containing q
end while
return failure

Algorithm 2 Adaptive leaf sweep and local refinement.

Input: LECT T, obstacle set $\mathcal{O}$, sweep budget
Output: Initial forest $\mathcal{F}$ and retained collision domains $\mathcal{C}$

classify priority cells by replaying or materializing LECT evidence
insert cells that pass validation into $\mathcal{F}$; prune covered descendants
retain inconclusive or blocked cells in $\mathcal{C}$
refine only selected domains in $\mathcal{C}$ using local materialization
insert refined boxes that pass validation, remain inside their domain, and connect to the forest
return $\mathcal{F}, \mathcal{C}$

C. Corridor Query and Certificate

After consolidation, G has policy-accepted vertices and edge types $(E_\cap, E_\varepsilon, E_s, E_o, E_\pi)$. Only $E_\cap$, E_o, and E_π are certified corridor or portal edges; E_ε, E_s, and other outputs require query validation.

Given start and goal configurations, the query locates containing boxes or configured anchor boxes and searches G. The certificate applies only to returned paths contained in conservative box regions or finite-cover OBB-corridor regions. Theorem 1 states the box-region case; Section III-C gives the finite-cover certificate for OBB-corridor edges. All other returns are query-validated results, not conservative box-corridor proofs.

Theorem 1 (Conditional conservative box-corridor path)**.** *Let $\mathcal{F}$ be a forest whose boxes have each passed conservative link-envelope validation against obstacle set $\mathcal{O}$. If a path $\gamma : [0,1] \to \mathcal{Q}$ is continuous and satisfies*

$$\gamma([0,1]) \subseteq \bigcup_{B \in \mathcal{F}} B,$$

then every configuration on γ is collision-free with respect to the collision model in Assumption 1.

Proof. For any $s \in [0,1]$, the containment assumption gives a conservatively validated box $B \in \mathcal{F}$ with $\gamma(s) \in B$. By construction, conservative validation of B means every conservative link envelope for B is disjoint from $\mathcal{O}$. Because each such envelope contains the link geometry for every configuration inside B (Corollary 1), the robot geometry at $\gamma(s)$ is also disjoint from $\mathcal{O}$. Since s was arbitrary, the whole path is collision-free. $\square$

Scope. The theorem establishes collision-freedom only for paths contained in conservatively validated regions. It does not establish coverage or completeness: missed paths can arise from finite depth or refinement budget, axis-aligned boxes around narrow passages, non-vanishing envelope overapproximation, or bridge and repair limits. Paths using uncovered space, residual non-corridor segments, smoothing, or boxes admitted by non-conservative filters are accepted only after independent query validation.

D. Dynamic Obstacle Updates

For obstacle edits, robot kinematics and LECT records stay fixed; only scene labels and graph edges change. Deletions may promote cached blocked domains after edited-scene validation. Insertions invalidate conflicting boxes and witnesses, then run bounded local refill. The update is local by design: it maintains the represented forest rather than rediscovering every newly collision-free region.

Algorithm 3 Obstacle deletion promotion and insertion refill.

Input: Forest $\mathcal{F}$, typed graph G, collision-domain cache $\mathcal{C}$, edit (Δ^-, Δ^+)
Output: Updated forest, typed graph, and collision-domain cache

```
if Δ⁻ ≠ ∅ then
    recheck cached blocked domains affected by deleted obstacles
    promote only domains that pass edited-scene validation and adjacency checks
end if
if Δ⁺ ≠ ∅ then
    remove boxes and typed witnesses hit by inserted obstacles
    run a bounded local leaf sweep inside each invalidated box
    insert replacement boxes that pass validation and connect to the surviving forest
end if
revalidate affected typed edges and update components
return updated (F, G, C)
```

After an update, retained or inserted boxes must satisfy the edited-scene validation policy, and typed edges must keep their original semantics. If the updated graph cannot provide a contained conservative corridor, the candidate path is accepted only after independent query validation.

VI. Experiments

The experiments evaluate this RBF implementation under saved scenes, stated validation policies, and audit-separated timing. The Shelf+IIWA profile study uses the KUKA LBR iiwa shelf-workcell benchmark and the manifest-registered RBF configuration; *registered* means fixed in the artifact manifest before table extraction. Planning rows use a common reporting convention for success, audit resolution, final simplification, online timing, and path-ratio references; offline checkpoint curves explicitly sweep planner budget. Dynamic-update rows are separate maintenance-only measurements of the scene cache. Random scenes and queries are saved before planner execution.

a) Protocol: Timings are wall-clock measurements on an Ubuntu 22.04 workstation with an Intel i5-12600KF CPU and 31 GiB memory. The generated tables and figures are tied to provenance manifests that record sources, thread and simplification policies, checksums, and artifact-integrity checks. The manifests record an 8-thread budget for RBF and LECT work and matching math-library thread limits; OMPL RRT-Connect, PRM, and BIT* are algorithmically single-threaded in these runners unless planner-internal parallelism is exposed by the backend. Table headers include a reusable-build term only for methods that construct a persistent object before queries: RBF, PRM, and the Shelf IRIS-NP/GCS row (nonlinear-programming IRIS followed by GCS optimization). RRT-Connect and BIT* are reported as per-query timing baselines without a reusable-build term.

Success requires query validation with the fixed 0.01-rad segment step. Online/q denotes per-query online time and excludes final simplification and audit time. When the optional per-query audit column (Audit/q) is present, audit-complete time is Online/q plus 0.010 s plus Audit/q. Numeric entries are medians with $[Q_1, Q_3]$ quartiles. Path-ratio columns use success-only audited query-level references unless a caption states otherwise; lower path ratios are better. The main tables summarize query-validated outcomes; they are not conservative-corridor certificates, repair-free results, or parameter-sensitivity studies. The reported runs check active rounded links against $\mathcal{O}$; self-collision, attached objects, grippers, and task-specific forbidden regions require additional scene checks before using the certificate.

b) Cache and profile boundary: Warm-cache rows include compatible LECT lookup and evidence reads. Snapshot creation, loading, resident memory, and offline setup amortization are outside those rows. Here d23 denotes the registered depth-23 LECT snapshot: 16,777,215 records, 12.75 GiB on disk, and 958.079 s prewarm time. Except for Live HIFK-5, Table II rows replay compatible d23 evidence; that row materializes HIFK-5 evidence live. The b100 label denotes the registered 100-box Shelf operating point. The baseline row uses the SupportHull representation with the standard AABB broadphase. Cache reuse requires matching robot model, endpoint source, link representation, split-depth descriptor, radii, and active-link set. Timing remains audit-separated. Bridge or repair witnesses become certificate-supported only after conversion into conservative box-corridor or OBB-corridor edges; all remaining witnesses stay query-validated transitions.

A. Shelf+IIWA Registered Profile Study

Fig. 3 shows the Shelf+IIWA scene and fixed five-anchor query cycle AS–TS–CS–LB–RB–AS; the anchor AS is the approach-side anchor. TS and CS are shelf-side target anchors, and LB and RB are the left and right bin anchors. Table II and Fig. 4 report the manifest-registered b100 profile over seeds 0–7. This fixed cycle is used for mechanism ablations and profile selection; the saved random

TABLE II
Shelf+IIWA RBF query-validated profile and mechanism ablations.

Case	Build	Online/q	Amort. 5q	$L/L^\star$
RBF baseline	0.166 [0.163, 0.170]	0.130 [0.104, 0.150]	0.164 [0.137, 0.183]	1.16 [1.08, 1.38]
Critical sample	0.282 [0.280, 0.285]	0.134 [0.103, 0.157]	0.190 [0.161, 0.213]	1.14 [1.08, 1.38]
Link AABB	0.141 [0.139, 0.143]	0.123 [0.099, 0.140]	0.151 [0.128, 0.168]	1.19 [1.08, 1.47]
Live HIFK-5	7.006 [6.947, 7.068]	0.234 [0.170, 0.278]	1.639 [1.576, 1.687]	1.14 [1.08, 1.39]
SupportHull only	0.349 [0.287, 0.363]	0.232 [0.155, 0.283]	0.296 [0.220, 0.352]	1.16 [1.08, 1.38]

Notes: Median $[Q_1, Q_3]$, s. Amort. 5q is Build/5 + Online/q; Online/q excludes final simplification and audit; $L/L^\star$ uses success-only query-level 0.01-rad audited references.

TABLE III
Shelf+IIWA audited-success per-query timing and path-quality comparison.

	RBF **b100** Build 0.166 s		**IRIS-NP/GCS** Build 355.425 s		**PRM** Build 20.040 s		**RRT-Connect** No reusable build		**BIT*** No reusable build	
Query	T (s)	$L/L^\star$	T (s)	$L/L^\star$	T (s)	$L/L^\star$	T (s)	$L/L^\star$	T (s)	$L/L^\star$
AS→TS	0.081 [0.078, 0.087]	1.39 [1.34, 1.48]	0.449 [0.449, 0.449]	1.44 [1.44, 1.44]	0.198 [0.147, 0.248]	1.61 [1.42, 1.73]	0.034 [0.024, 0.052]	2.27 [2.05, 2.72]	4.004 [4.002, 4.004]	1.06 [1.02, 1.08]
TS→CS	0.163 [0.158, 0.170]	1.12 [1.09, 1.14]	0.269 [0.269, 0.269]	1.26 [1.26, 1.26]	0.173 [0.118, 0.312]	2.99 [2.11, 3.22]	0.054 [0.021, 0.089]	3.98 [2.80, 5.09]	10.138 [10.004, 10.386]	1.09 [1.04, 1.11]
CS→LB	0.109 [0.104, 0.118]	1.47 [1.32, 1.82]	0.381 [0.381, 0.381]	1.07 [1.07, 1.07]	0.136 [0.120, 0.195]	1.49 [1.39, 2.09]	0.032 [0.027, 0.070]	2.20 [1.75, 2.51]	7.504 [7.504, 7.725]	1.03 [1.01, 1.06]
LB→RB	0.132 [0.128, 0.141]	1.15 [1.07, 1.25]	0.786 [0.786, 0.786]	1.07 [1.07, 1.07]	0.164 [0.091, 0.399]	1.18 [1.09, 1.27]	0.001 [0.001, 0.002]	1.29 [1.27, 1.32]	1.654 [1.654, 1.654]	1.02 [1.02, 1.03]
RB→AS	0.143 [0.138, 0.150]	1.04 [1.03, 1.05]	0.527 [0.527, 0.527]	1.01 [1.01, 1.01]	0.252 [0.096, 0.295]	1.23 [1.17, 1.24]	0.001 [0.001, 0.001]	1.07 [1.03, 1.10]	0.204 [0.204, 0.204]	1.02 [1.01, 1.03]

Notes: Median $[Q_1, Q_3]$, s. T = Online/q, excluding final simplification and audit; $L/L^\star$ uses success-only query-level 0.01-rad audited reference paths. AS, TS, CS, LB, and RB denote the approach-side anchor, two shelf-side anchors, and the left and right bin anchors.

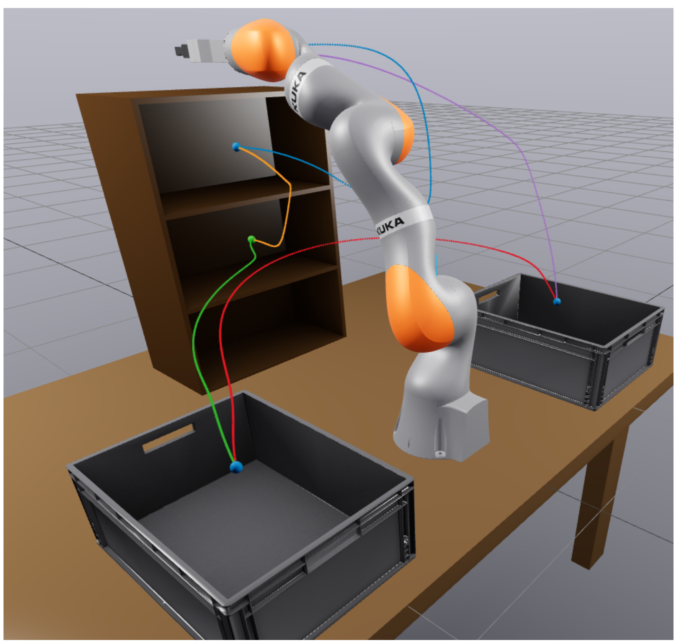


Fig. 3. Canonical Marcucci Shelf+IIWA scene with the fixed saved five-anchor query cycle.

catalog below tests broader start-goal variation. The table reports build time, Online/q, five-query amortized time Build/5 + Online/q, and path ratio. The Critical sample row uses the same planner and fixed-step query validation; its endpoint approximation is treated as a non-certifying filter rather than as a conservative endpoint certificate.

The registered RBF baseline uses d23 SupportHull: Build 0.166 s, Online/q 0.130 s, and Amort. 5q 0.164 s; Table II gives quartiles. The Live HIFK-5 row has a 7.006 s live-evidence build; the other rows replay d23 evidence.

After corridor conversion, the Shelf manifest records no residual ordinary (non-corridor) segments for the 40 OBB-corridor paths in Table II. The representation rows compare cost: Link AABB lowers Online/q from 0.130 s to 0.123 s at a similar path ratio; SupportHull-only raises five-query time from 0.164 s to 0.296 s.

As a validation-boundary diagnostic, the manifest-registered query pipeline produced 40/40 audited paths. An ablated variant with query bridge, repair, shortcutting, and final simplification disabled produced 0/40; this compound ablation shows that accepted Shelf paths depend on those auxiliary stages, but it does not isolate single-stage effects. The Shelf rows in Table II are query-validated results, not conservative-corridor certificates.

B. Shelf Cross-Algorithm Context

The cross-algorithm shelf study uses the same five queries, 0.01-rad audit segment step, and 0.010 s final-simplification budget for RBF, IRIS-NP/GCS, PRM, RRT-Connect, and BIT*. RBF uses the b100 profile and per-query Online/q; PRM and IRIS-NP/GCS report reusable builds, while RRT-Connect and BIT* are single-query rows with no reusable-build term. BIT* checkpoints are offline quality–time references, not online stopping rules.

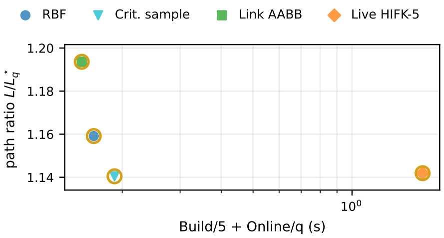


Fig. 4. Shelf+IIWA RBF mechanism trade-off. Query-validated mechanism rows are plotted at Build/5 + Online/q versus $L/L^\star_{\mathrm{q}}$; gold rings mark the rows listed in Table II.

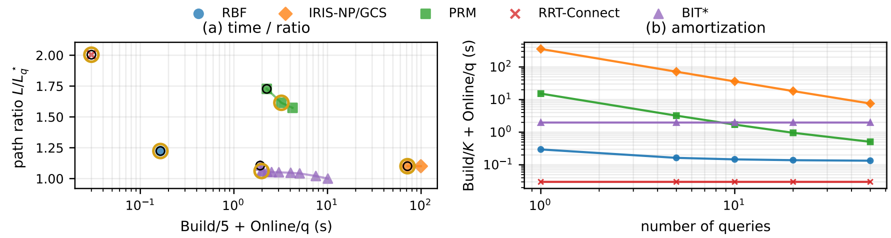


Fig. 5. Shelf+IIWA cross-algorithm trade-off. Black rings mark the first checkpoints with audited success on the full saved query set. Timings use the Online/q column of Table III; reusable-build terms are added only for methods with a reusable-build measurement.

TABLE IV
Saved-catalog random-scene operating points.

Scenario	RBF			PRM			RRT-Connect		BIT*	
	Build	Online/q	$L/L^\star_{\mathrm{q}}$	Build	Online/q	$L/L^\star_{\mathrm{q}}$	Online/q	$L/L^\star_{\mathrm{q}}$	Online/q	$L/L^\star_{\mathrm{q}}$
IIWA-Medium	0.075 [0.058, 0.083]	0.009 [0.007, 0.011]	1.00 [1.00, 1.03]	3.567 [3.007, 3.933]	0.038 [0.014, 0.070]	1.03 [1.02, 1.12]	0.002 [0.001, 0.002]	1.02 [1.01, 1.05]	0.107 [0.087, 0.154]	1.00 [1.00, 1.01]
IIWA-Hard	0.097 [0.086, 0.105]	0.014 [0.011, 0.015]	1.00 [1.00, 1.03]	4.175 [3.408, 5.637]	0.048 [0.021, 0.102]	1.03 [1.02, 1.08]	0.002 [0.002, 0.003]	1.03 [1.00, 1.07]	0.129 [0.097, 0.186]	1.00 [1.00, 1.01]
UR5-Medium	0.262 [0.254, 0.270]	0.075 [0.020, 0.086]	1.05 [1.00, 1.14]	3.124 [2.966, 3.418]	0.323 [0.085, 0.426]	1.02 [1.00, 1.12]	0.023 [0.002, 0.049]	1.23 [1.06, 1.54]	0.134 [0.101, 0.167]	1.01 [1.00, 1.08]
UR5-Hard	0.276 [0.264, 0.284]	0.083 [0.043, 0.108]	1.05 [1.01, 1.14]	2.488 [2.325, 3.048]	0.419 [0.129, 0.437]	1.06 [1.01, 1.19]	0.048 [0.008, 0.090]	1.36 [1.14, 1.58]	0.154 [0.115, 0.171]	1.03 [1.00, 1.13]
Panda-Medium	0.250 [0.218, 0.265]	0.028 [0.013, 0.039]	1.05 [1.02, 1.09]	3.431 [3.364, 4.180]	0.066 [0.038, 0.159]	1.06 [1.03, 1.08]	0.005 [0.002, 0.011]	1.22 [1.12, 1.35]	0.116 [0.092, 0.152]	1.00 [1.00, 1.00]
Panda-Hard	0.224 [0.221, 0.237]	0.043 [0.021, 0.053]	1.05 [1.01, 1.16]	2.337 [1.627, 3.978]	0.054 [0.034, 0.126]	1.10 [1.05, 1.16]	0.056 [0.019, 0.136]	1.25 [1.14, 1.37]	0.164 [0.123, 0.214]	1.01 [1.00, 1.09]

Notes: Median $[Q_1, Q_3]$, s. Build is a per-scene reusable-build term for RBF and PRM; RRT-Connect and BIT* are single-query rows. Online/q excludes final simplification and audit; $L/L^\star_{\mathrm{q}}$ uses success-only saved 0.01-rad audited query-level references.

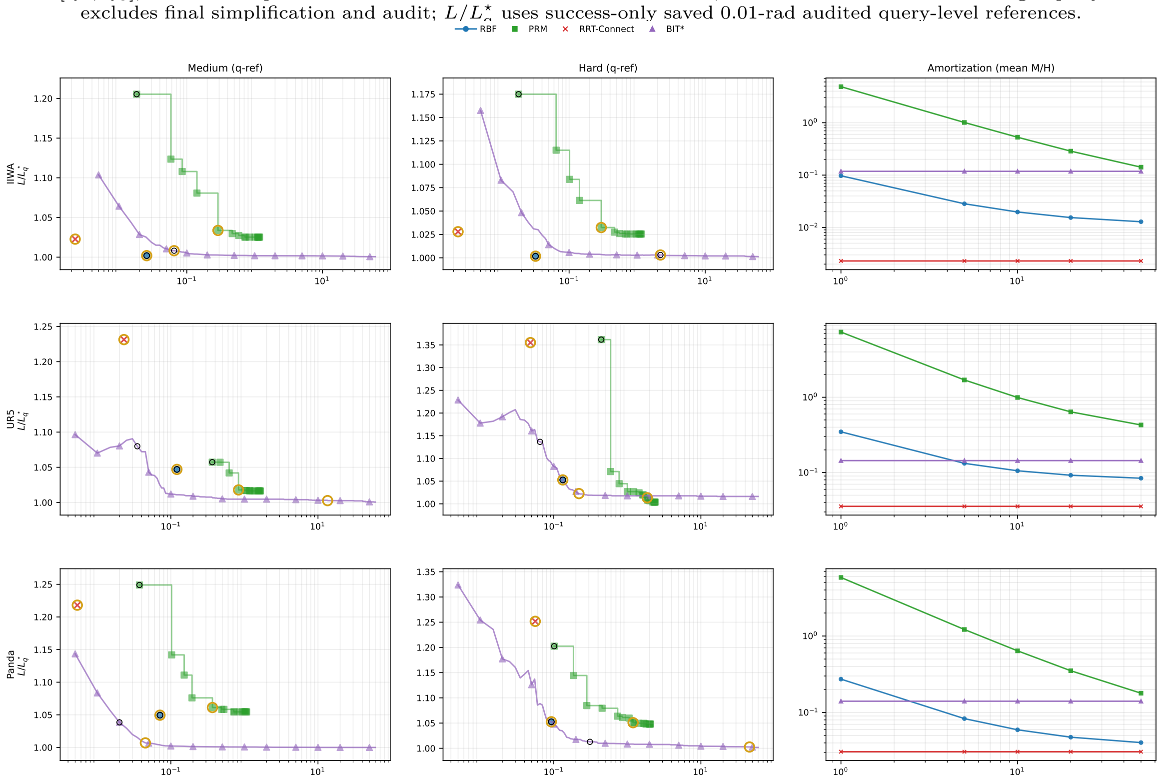


Fig. 6. Saved-catalog random-scene trade-off. Left and middle panels show a fixed five-query amortization, Build/5 + Online/q, for medium and hard rows; the right panel varies the amortization count K in Build/K + Online/q. Rings mark the first checkpoints with audited success on the full saved query set (black) and later $\geq 1\%$ path-ratio improvements (gold); PRM and BIT* use offline checkpoints.

Under these definitions, reusable builds are 0.166 s (RBF), 20.040 s (PRM), and 355.425 s (IRIS-NP/GCS), excluding RBF d23 prewarm, snapshot load, and resident memory. For all five queries, RRT-Connect has lower median T and higher $L/L^\star$ than RBF, while BIT* has higher median T and lower $L/L^\star$.

TABLE V
Endpoint-envelope source comparison over fixed joint-box widths.

	Volume ratio					Time (μs)					Gap (mm)				
Source	0.02	0.10	0.20	0.30	0.50	0.02	0.10	0.20	0.30	0.50	0.02	0.10	0.20	0.30	0.50
IFK-AA	1.09	1.52	2.30	3.53	8.61	5.48	6.45	6.58	6.81	7.35	–	–	–	–	–
HIFK-3	1.04	1.22	1.51	1.89	3.12	25.94	29.16	27.21	29.27	29.38	–	–	–	–	–
HIFK-5	1.02	1.13	1.30	1.51	2.07	70.40	73.82	73.15	75.92	72.32	–	–	–	–	–
Critical sample	1.00	1.00	1.00	0.99	0.98	8.68	10.03	11.56	12.69	16.33	-0.05	-1.17	-4.53	-10.77	-30.97
Analytical	1.00	1.00	1.00	1.00	1.00	6429.96	6602.25	6800.10	6927.12	7096.01	0.00	0.00	0.00	0.00	-4.49
MC	0.64	0.74	0.78	0.79	0.82	2251.74	11233.26	22352.95	33813.58	56835.57	-6.05	-32.68	-65.57	-88.87	-123.53

Notes: Volumes are normalized to Analytical at each width. IFK and HIFK are the certificate-backed rows; Critical sample, Analytical, and Monte Carlo (MC) are diagnostic or reference rows. Gap is the sampled-union signed gap (mm); dashes mark rows where the diagnostic is omitted.

TABLE VI
Link-envelope representation comparison over fixed joint-box widths.

	Volume						Build (μs)						Test (μs)					
Envelope	0.02	0.05	0.10	0.20	0.30	0.50	0.02	0.05	0.10	0.20	0.30	0.50	0.02	0.05	0.10	0.20	0.30	0.50
Link AABB	0.0136	0.0201	0.0384	0.147	0.508	4.61	0.138	0.152	0.167	0.166	0.171	0.178	0.586	0.605	0.601	0.604	0.630	0.790
SupportHull	0.000359	0.00266	0.0143	0.106	0.449	4.54	0.205	0.214	0.218	0.219	0.217	0.250	12.520	12.850	12.267	12.223	12.659	13.026

Notes: Means over fixed-width boxes. Test time pools far-separated probes and local obstacle-overlap probes; build time excludes endpoint time.

C. Random Multi-Robot Scenes

The random-scene study uses saved prefix catalogs: ten pre-generated queries per scene reused by every method. Across KUKA LBR iiwa, UR5, and Panda, medium and hard rows over ten seeds give 600 saved query evaluations per method. For each robot and seed, medium and hard rows share the saved start-goal set; hard rows add later obstacle prefixes instead of regenerating unrelated scenes. Fixed-audit probes exclude direct-line cases before table extraction. The trade-off plot uses Build/5 + Online/q as a fixed amortization reference, while the saved catalogs still contain ten queries per scene and the right panel varies the amortization count K. Table IV and Fig. 6 summarize the operating points and corresponding checkpoint curves. RBF builds one query-agnostic forest per scene and reuses it for ten queries; random rows use LECT caches and anchors tied to each robot rather than Shelf-specific assets. Each RBF row in Table IV uses the OBB-corridor profile and passes the fixed audit on the full saved query set. Repair witnesses contribute to accepted paths only after conversion into conservative C-space OBB-corridor regions; the generated summaries list no residual ordinary (non-corridor) segments for those RBF rows. PRM is built once per scene, whereas RRT-Connect and BIT* solve each query independently. The archived IRIS-NP/GCS saved-catalog run is retained only as a feasibility diagnostic because its saved-query audit outcomes did not support comparable timing and path-quality rows: hard rows had 0/100 successes for each robot, and medium rows had 0/100 (IIWA), 30/100 (UR5), and 10/100 (Panda) successes. The comparison therefore focuses on online latency and path quality under shared catalogs. RRT-Connect has the lowest median Online/q in five of six operating points; the exception is Panda-Hard, where RBF has the lower median Online/q. RRT-Connect has higher median path ratios than RBF in all six. With the scene forest reused, RBF has lower median Online/q than BIT* in all six cases; $L/L^\star_{\mathrm{q}}$ matches BIT* to table precision for IIWA and is worse for UR5/Panda.

D. Mechanism Studies

Tables V and VI compare endpoint sources and envelope narrow phases. Table V reports endpoint volume as a ratio to the Analytical row at the same width and the sampled-union signed gap in millimeters; negative gaps are finite-sampling diagnostics. Only IFK and HIFK rows are certificate-backed; Critical sample, Analytical, and MC rows are diagnostic or reference rows rather than certificate evidence. Table VI motivates the AABB broadphase: SupportHull/GJK stores a compact support record, but wide boxes increase collision-test time when every obstacle reaches the GJK narrow phase.

E. Dynamic Updates

Table VII reports scene-cache maintenance only: no query bridge, connector, simplification, audit, or query timing is included.

TABLE VII
Adaptive leaf-sweep maintenance-only timing.

Operation	Time (ms)	Speedup over 15-obstacle build
Warm build (10 obstacles)	922.75 [897.57, 955.67]	–
Insert to 15 obstacles	6.06 [5.69, 6.54]	153.3–207.5×
Remove from 15 obstacles	0.35 [0.31, 1.65]	943.6–3354.0×
Warm build (15 obstacles)	958.79 [941.67, 1115.18]	–

Notes: Median $[Q_1, Q_3]$, ms. Maintenance only; no query stage. Speedup compares each update row with the 15-obstacle warm build.

A diagnostic audit on locally updated forests and matching freshly rebuilt forests, each without the query bridge and repair stages, produced 0/8 paths in both cases. The 0/8 result is a scope check, not evidence of post-edit query success; Table VII reports local maintenance timing only, not timing for a complete pipeline that updates the scene and solves a query.

VII. Conclusion

RBF is a link-envelope-guided box-forest planner for multi-query manipulation. It uses axis-aligned boxes as the reusable region type: endpoint-to-link swept-capsule bounds provide the validation test, policy-accepted boxes form a scene-level forest and typed corridor graph, and LECT separates reusable kinematics-keyed evidence from scene labels. Certificates apply only to conservative box unions or OBB-corridor unions; repaired, shortened, and post-processed paths require independent query validation.

Under saved catalogs, audit-separated timing, and explicit cache accounting, warm-cache depth-23 LECT replay on Shelf+IIWA yields a 0.166 s reusable build and 0.130 s Online/q, whereas live HIFK-5 materialization takes 7.006 s. In saved random scenes, RBF has lower Online/q than BIT* at all six operating points in Table IV, while BIT* has lower, hence better, path ratios in the four UR5/Panda cases. The dynamic-update rows measure scene-cache maintenance rather than a complete pipeline that updates the scene and solves a query. Within this scope, RBF is a low-overhead method for constructing reusable regions for repeated queries. Remaining limitations are the lack of certificate coverage for repaired, shortened, and post-processed paths; the absence of a complete update-and-query timing pipeline; unreported LECT snapshot-load and memory break-even analyses; and validation beyond active-link obstacle collision, including self-collision, payload and gripper models, and task-specific regions.

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